StatefulSet + Amazon EKS Real World Setup Guide

1. Real World Architecture Overview

```
                AWS Region
                     |
        ---------------------------------
        |               |               |
      AZ-1a           AZ-1b           AZ-1c
        |               |               |
    Worker-1        Worker-2        Worker-3
        |               |               |
      Pod-0           Pod-1           Pod-2
        |               |               |
      PVC-0           PVC-1           PVC-2
        |               |               |
      EBS-0           EBS-1           EBS-2

Each replica in StatefulSet gets its own EBS volume.
```

2. Prerequisites

```
- EKS Cluster created
- Worker nodes in multiple AZs
- IAM OIDC provider enabled
- AWS EBS CSI Driver installed
```

3. Install AWS EBS CSI Driver (EKS Add-on)

```
aws eks create-addon   --cluster-name my-cluster   --addon-name aws-ebs-csi-driver
```

4. Create StorageClass

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: ebs-sc
provisioner: ebs.csi.aws.com
volumeBindingMode: WaitForFirstConsumer
parameters:
  type: gp3
  encrypted: "true"
```

5. StatefulSet Example (MySQL)

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: mysql
```

```yaml
spec:
  serviceName: mysql
  replicas: 3
  selector:
    matchLabels:
      app: mysql
  template:
    metadata:
      labels:
        app: mysql
    spec:
      containers:
      - name: mysql
        image: mysql:8
        volumeMounts:
        - name: mysql-data
          mountPath: /var/lib/mysql
  volumeClaimTemplates:
  - metadata:
      name: mysql-data
    spec:
      accessModes: ["ReadWriteOnce"]
      storageClassName: ebs-sc
      resources:
        requests:
          storage: 20Gi
```

6. Production Best Practices

```
- Use Multi-AZ worker nodes
- Use WaitForFirstConsumer binding mode
- Enable EBS encryption
- Take regular EBS snapshots
- Use PodDisruptionBudgets
- Monitor using CloudWatch + Prometheus
```